

EGD Project

Develop a data engineering project to obtain, analyze and process a dataset of your choice in order to find relevant information and patterns from the data, or implement a ML pipeline for some predictive purpose.

DATASETS

Tech

- <https://www.netresec.com/?page=PcapFiles>
- <https://dataverse.harvard.edu> (e.g. Online Shopping Store - Web Server Logs)
- <https://www.unb.ca/cic/datasets/index.html>
- <https://www.stratosphereips.org/datasets-overview>
- <https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=508099>

Business

- <http://insideairbnb.com/get-the-data/>
- USA flight data: [www.kaggle.com/ usdot/flight-delays/data](http://www.kaggle.com/usdot/flight-delays/data)

Other sources

<https://huggingface.co/datasets>

<https://developer.imdb.com/non-commercial-datasets/>

<https://www.kaggle.com/datasets>

<https://data.gov/>

or any topic you like, have data, and makes sense for this course.

OUTCOMES

Write a short report with results and analysis of up to 6 pages, using the format available in Moodle.

The report should consider the following topics:

- Introduction (problem description);
- Related work;
- Dataset profiling;
- Queries or/and Learning pipeline;
- Runtime/performance using different number of computing nodes;
- Results analysis.

Spark Platform

For the execution platform use the Google Dataproc Spark engine / Colab / Local. Dataproc can be used after registration, for \$300 USD or 3 months of free access: (<https://cloud.google.com/dataproc>).

To be presented on the lecture of: 22/05/2026

To be delivered until: 29/05/2026